

Postprocessing

Computer Graphics
Instructor: Sungkil Lee

Ambient Occlusion

Ambient Occlusion (AO)

- **Visibility-only approximate evaluation of rendering equation:**

$$A_{\bar{p}} = \frac{1}{\pi} \int_{\Omega} V_{\bar{p}, \hat{\omega}} (\hat{n} \cdot \hat{\omega}) \, d\omega$$

- A measure of how much such a point is hidden by its surroundings

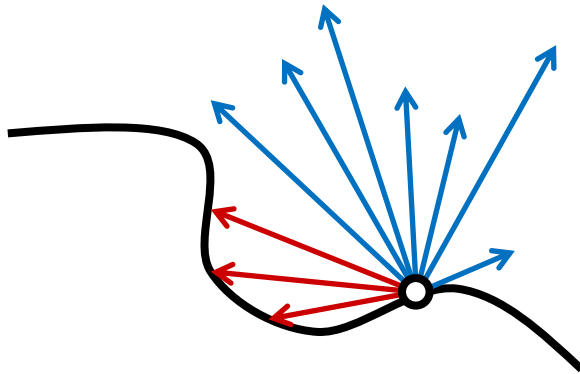
Screenshots from “Skyrim”



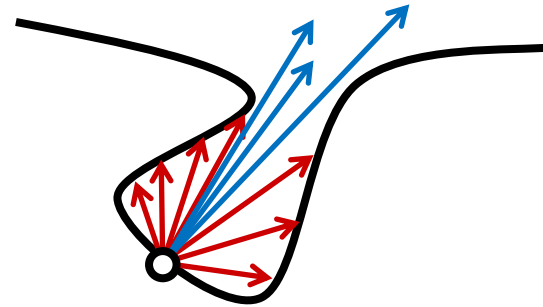
Object-Space Ambient Occlusion

- **Basic idea**

- Compute an occlusion factor for each point on a surface
- Incorporate this into the ambient term in shading models



Little occlusion: bright



Mostly occluded: dark

Screen-Space Ambient Occlusion (SSAO)

- **Limitations of object-space AO**

- Offline renderers typically do it by casting a large number of rays
- Impractical for real-time rendering and dynamic scenes

- **Screen-Space Ambient Occlusion [Mittring 2007]**

- In 2007, Crytek introduced a real-time solution for Crysis.
- SSAO is an approximation to the object-space ambient occlusion.
 - Based solely on screen-space depths.

GameStudio A7



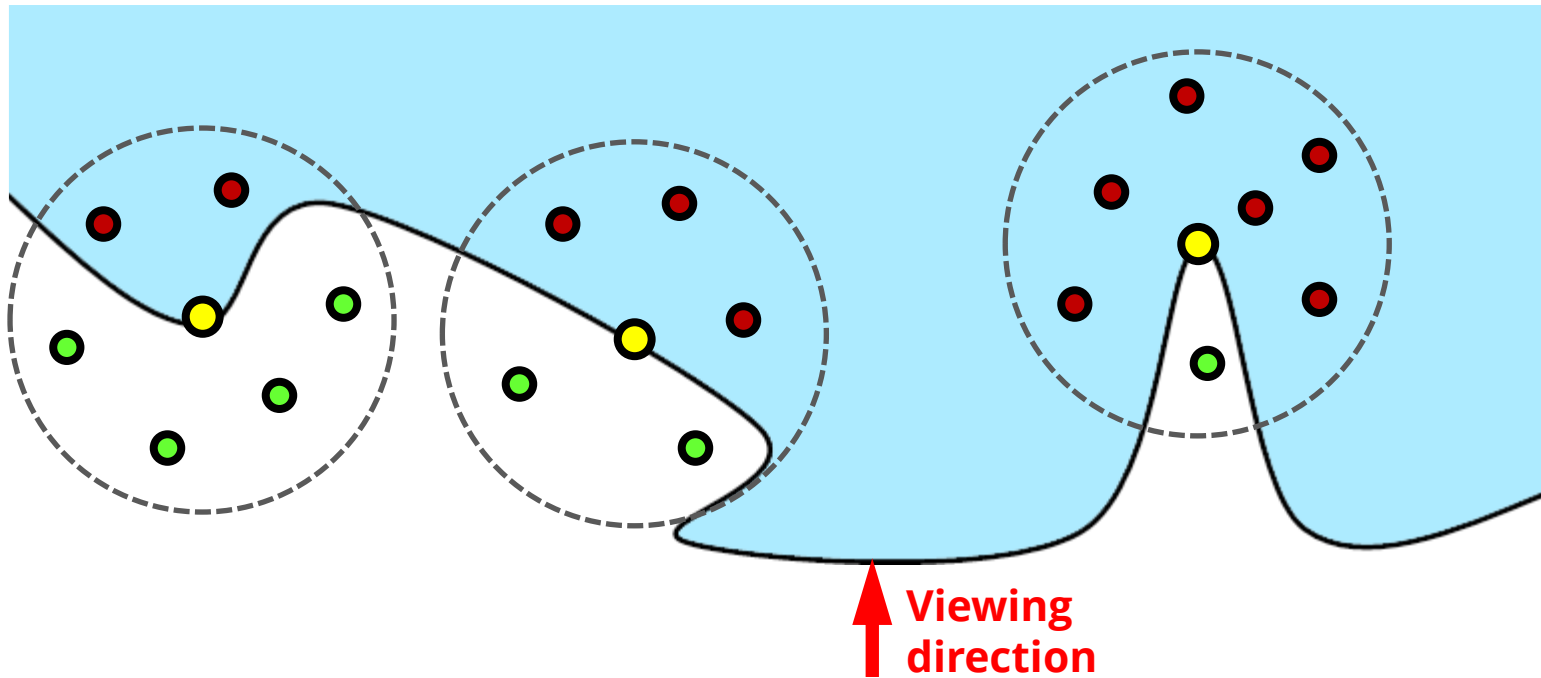
SSAO off

SSAO on

Screen-Space Ambient Occlusion (SSAO)

- **Basic idea**

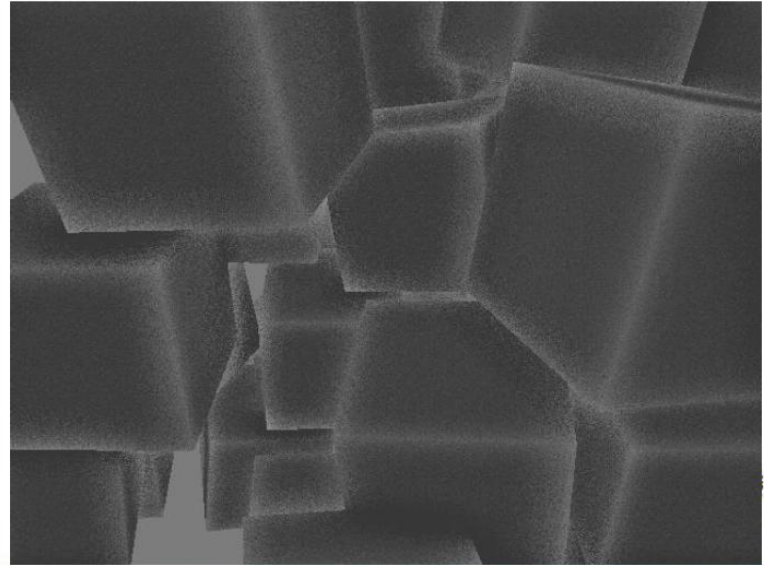
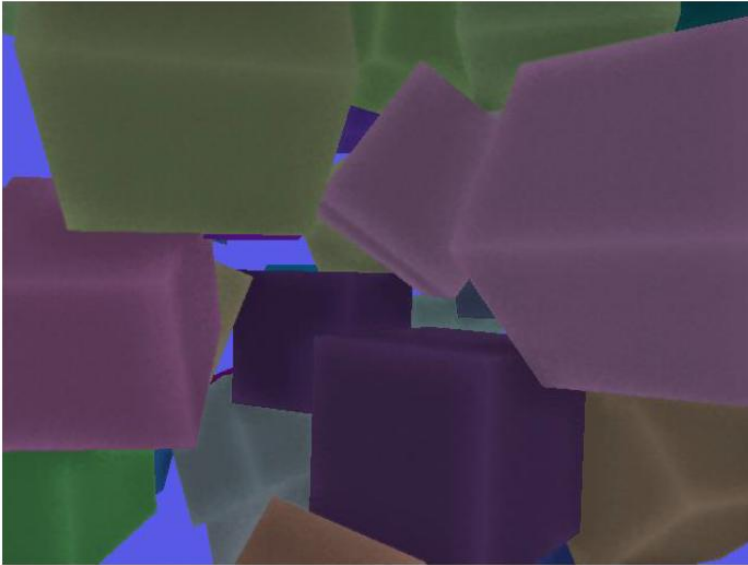
- Take samples in the sphere around the point.
 - Z component is a depth offset
- If sample + depth offset is smaller than the center pixel depth



Screen-Space Ambient Occlusion (SSAOO)

- **Limitations**

- Outer edges become brighter
- Limited radius in screen space



[Hoef 2012]

Motion Blur

Motion Blur

- **Informal definition**

- Illusion of rapidly moving objects blurred out having directional components.

- **Benefit**

- Important for sense of speed and direction



Split second motion blur, Ritchie, SIGGRAPH 2001 Course.

Physics

- **Underlying physics:**

- Aliasing from under-sampling along time
 - result from discrete sampling instead of continuous exposure in camera
 - So, it's often called the *temporal antialiasing*.



Shutter speed 1/200 s

<http://www.beckimpressionsphotography.com/photography-101-shutter-speed-motion-tutorial/>



Shutter speed 1/30 s



Shutter speed 1/2000 s



Shutter speed 1/8 s

<https://franklinll.files.wordpress.com/2014/04/shutter-speed.jpg>

References

- **Distributed ray tracing [Cook 1984]**

- A straightforward way to achieve is to collect more time samples and blend them in a single image.
 - Distributed ray tracing or accumulation buffering can be used.

- **Accumulation buffering [Haeberli 1990]**

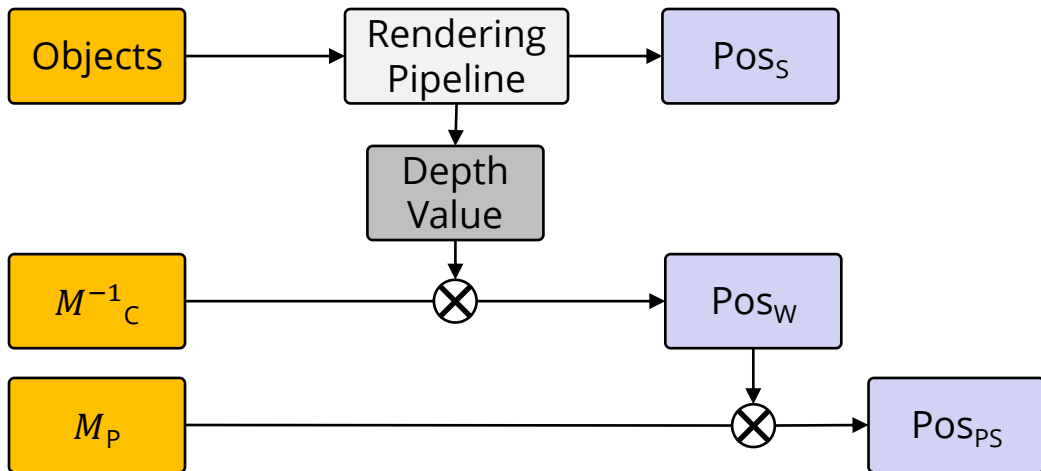
- Straightforward extension of distributed ray tracing to raster pipelines
- Render the scene as many as different time samples, and blend them!



Post-processing

• First pass: Motion-vector field rendering

- Computing the difference in the viewport position between the current frame and the previous frame
- Either of screen-space or object-space motions can be used.
- Compute world space positions for each pixel by using the depth value.

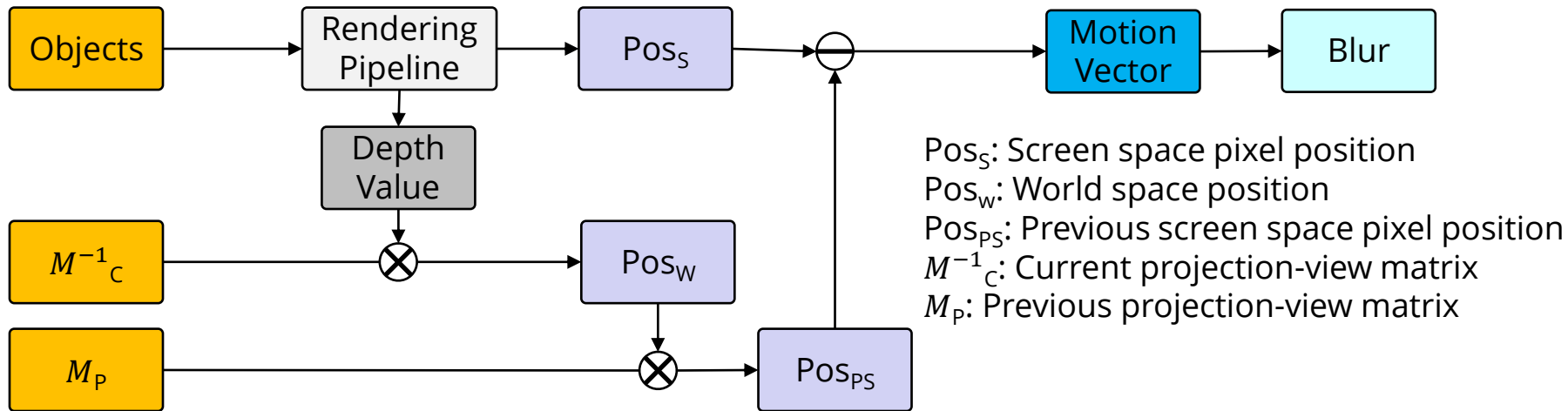


Pos_S: Screen space pixel position
Pos_W: World space position
Pos_{PS}: Previous screen space pixel position
 M^{-1}_C : Current projection-view matrix
 M_P : Previous projection-view matrix

Post-processing

- **Second pass: Directional line blur along the motion vector**

- Gather multiple samples along the motion direction.
- Averaging them out along the way to generate a blur.



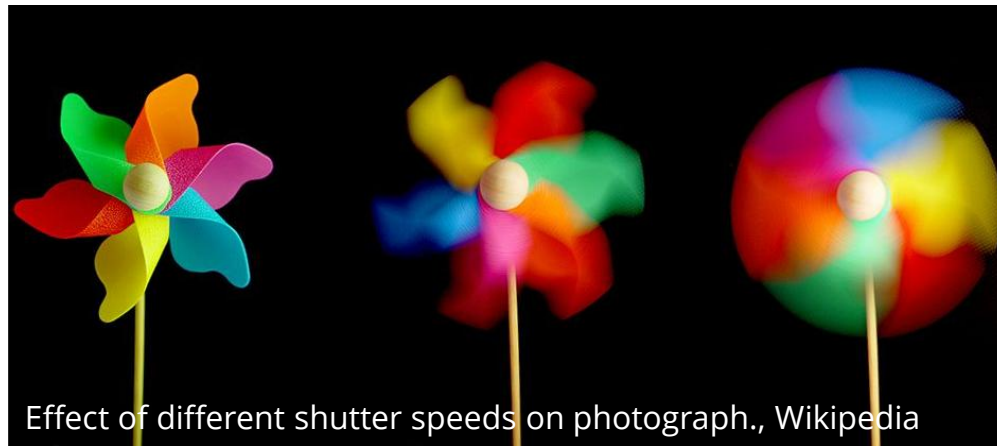
Limitations

- **Wrong visibilities**

- Correct visibilities along time are not considered.
- **Objects visibilities** are not taken into account (usually, camera-motion only)
- A better way should **spreading/splatting** of source pixels rather than the gathering of neighboring pixels.

- **Rigid-body linear blur only**

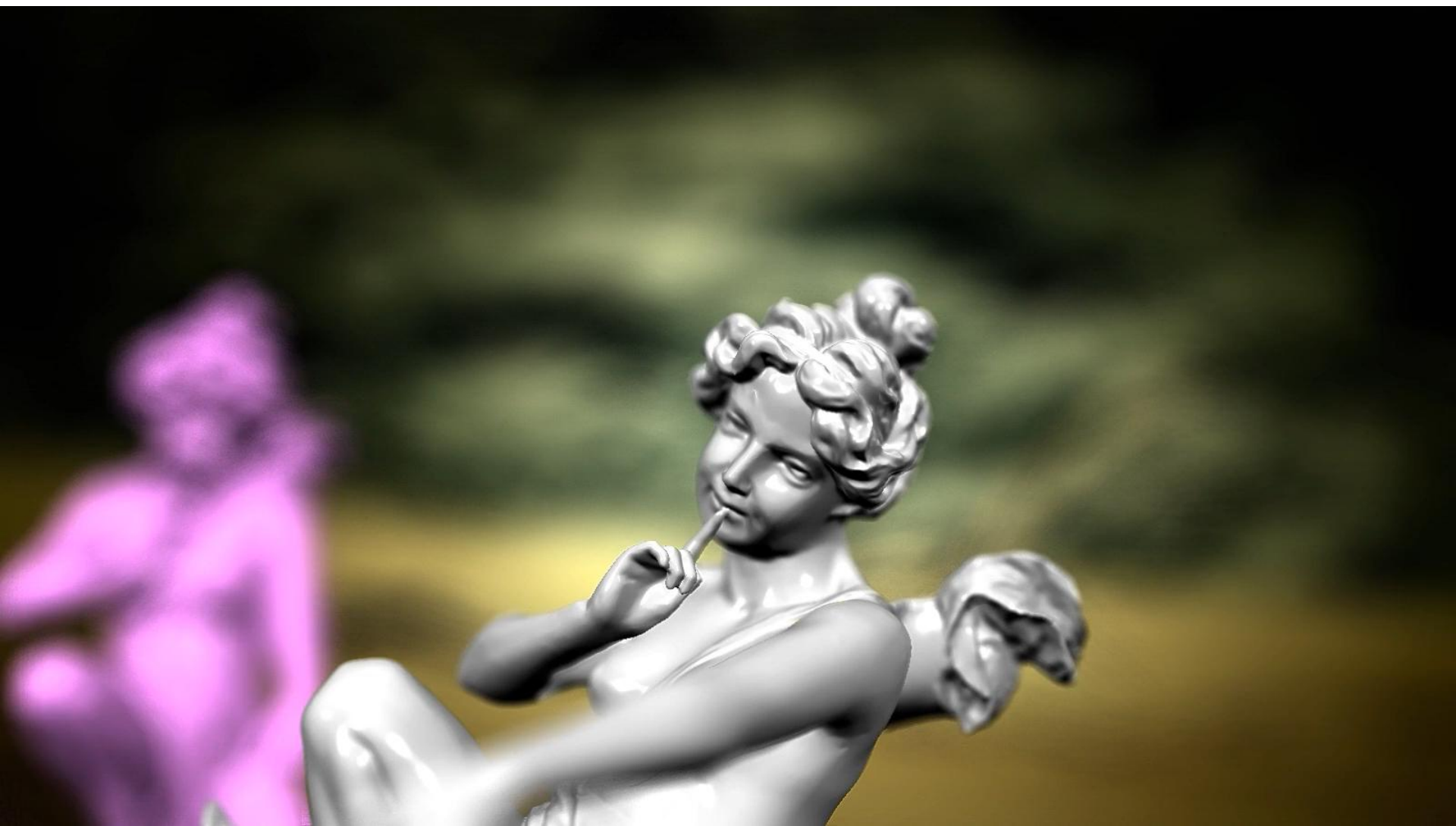
- Deformable objects cannot be handle, but approximation would be fine.
- Non-linear blur (e.g., rotational blur) needs radial blur.



Defocus Blur

Examples

- **Depth-of-Field Rendering with Multiview Synthesis (Lee et al., SIGGRAPH ASIA 2009)**



Examples

- **Real-Time Dynamic Bokeh Rendering with Efficient Look-Up Table Sampling (Jeong et al., IEEE TVCG 2022)**



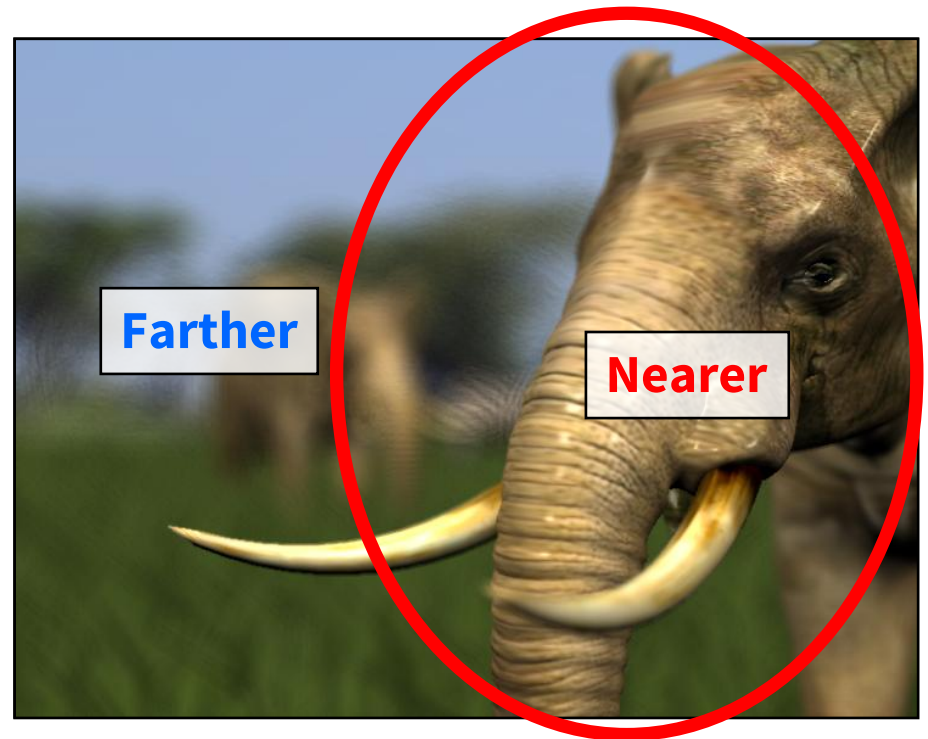
Depth of Field

- **Definition**

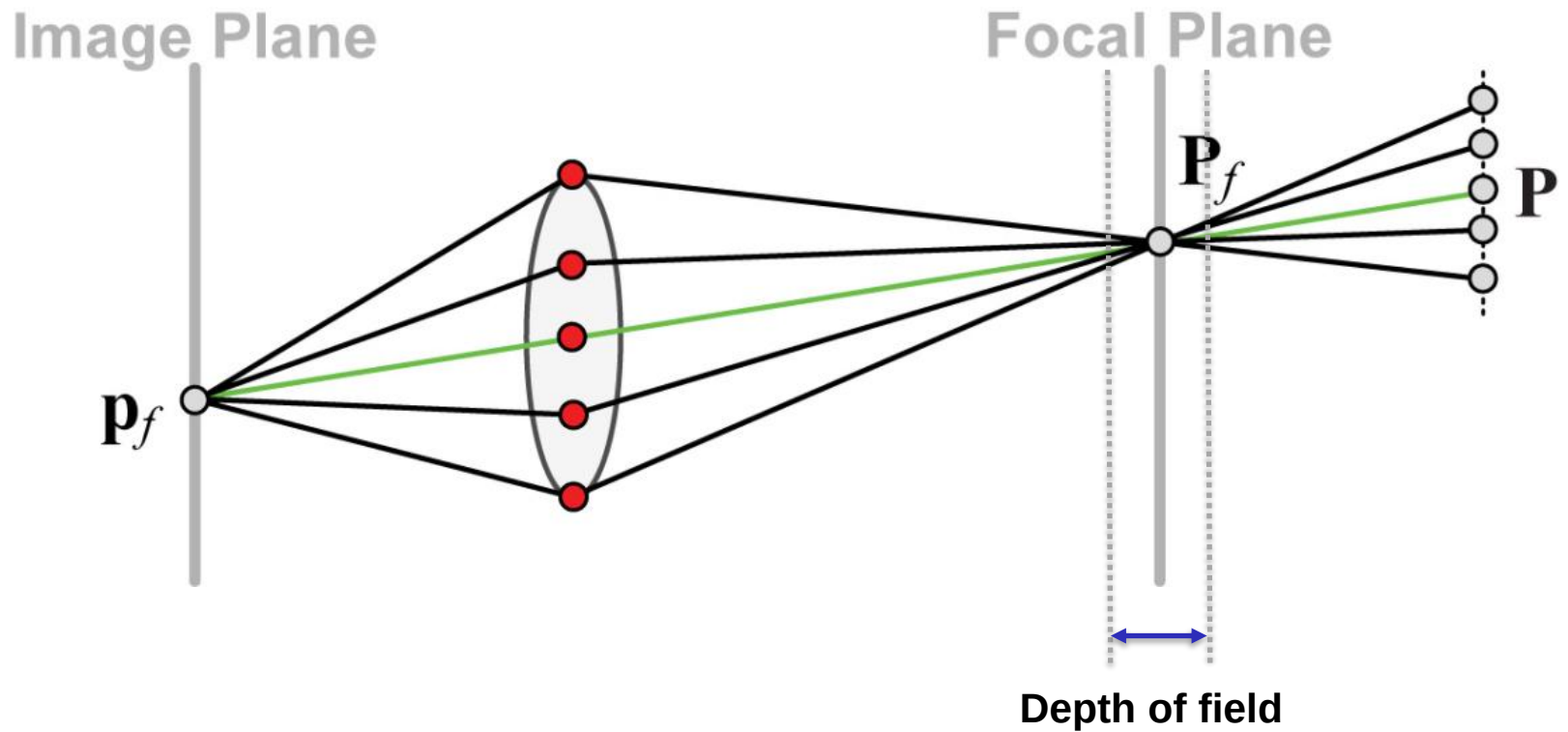
- Distance between the nearest and farthest objects in a scene that appear acceptably sharp in an image.

- **Benefits**

- Photorealism
- Monocular depth cue
- Attract attention/guide observer

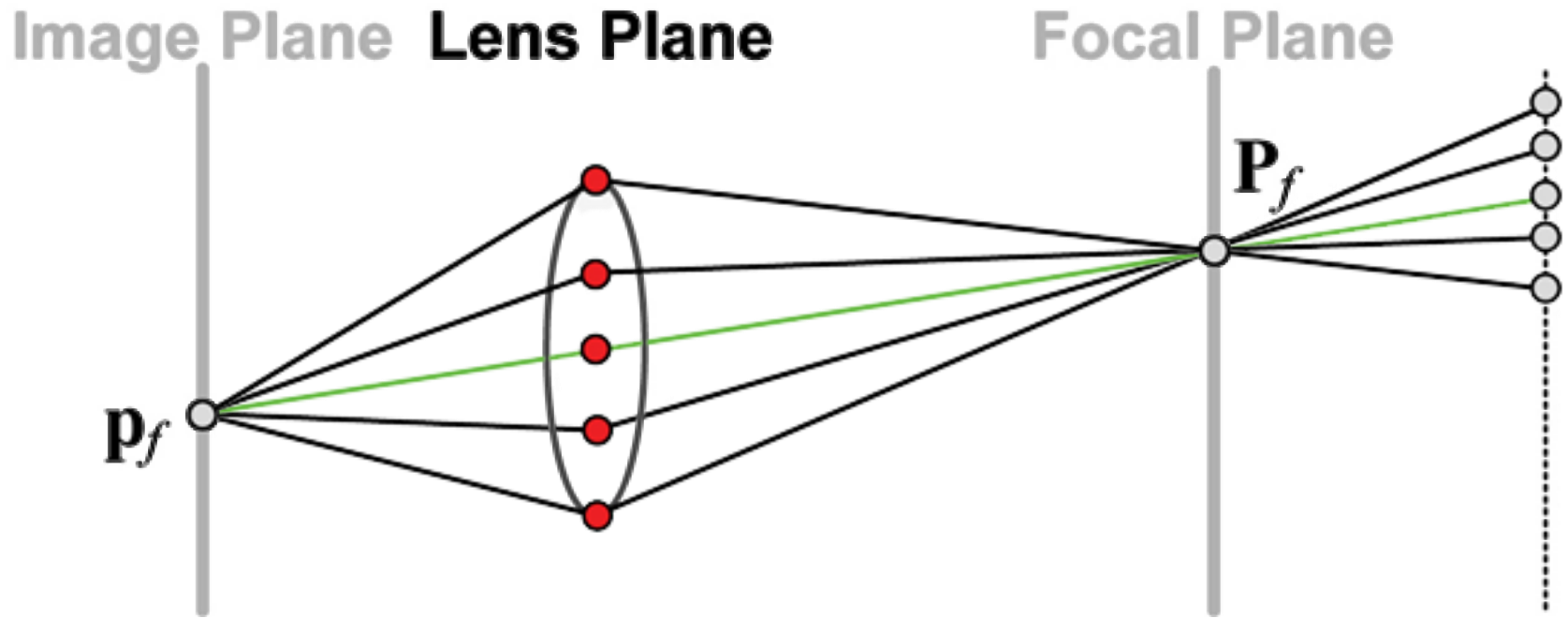


Underlying Physics



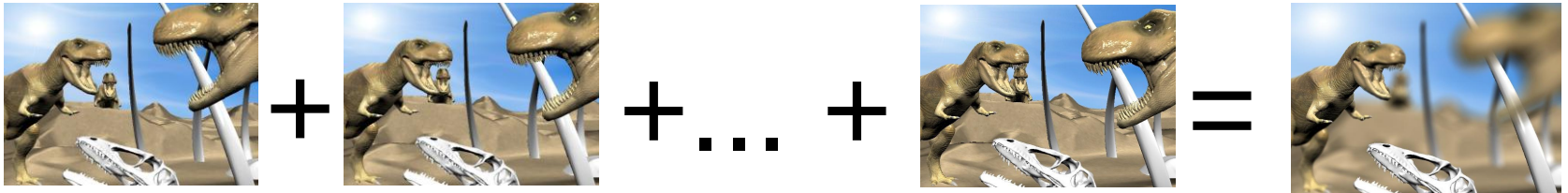
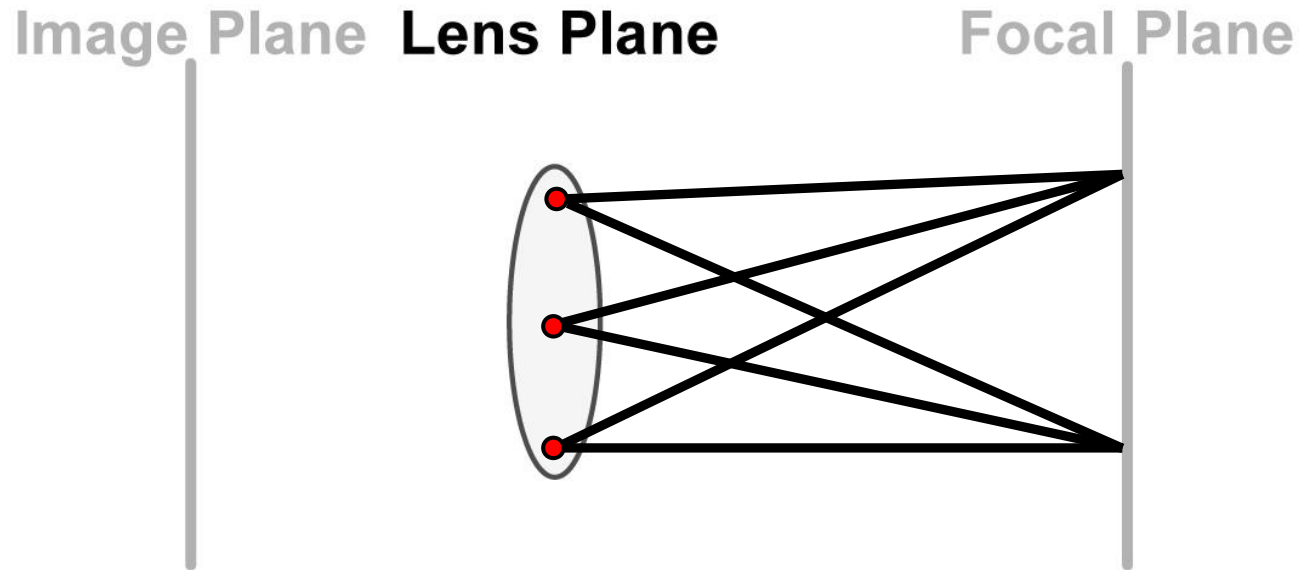
References: Object-Based Approaches

- Distributed ray tracing [Cook 1984]



References: Object-Based Approaches

- **Accumulation buffer method [Haeberli 1990]**



References: Object-Based Approaches

- **Pros**

- Produce physically-faithful results.

- **Cons**

- Costly rendering of 3D models
- Over-rasterization

Image-Based Approaches

- Image space ray tracing with depth layers [Lee 2010]

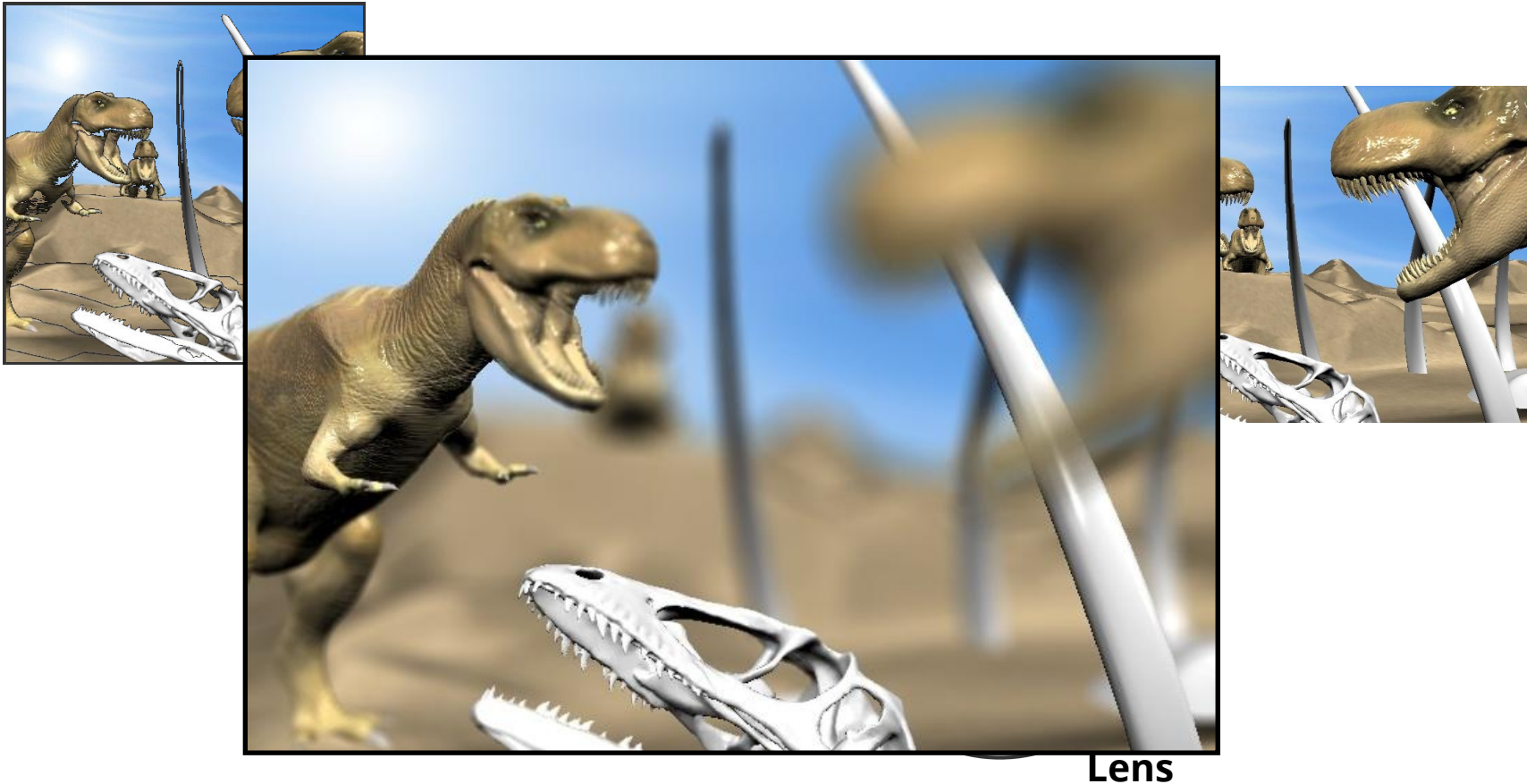
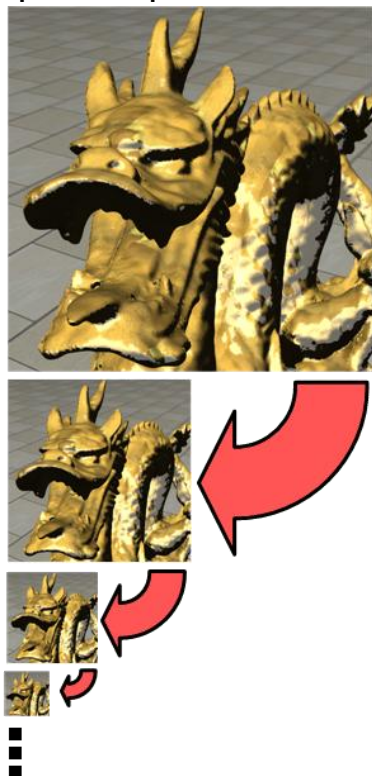


Image-Based Approaches

- **Post-filtering of a single-view image using a depth map**
 - Common in modern games

Mipmap interpolation [Lee 2009]



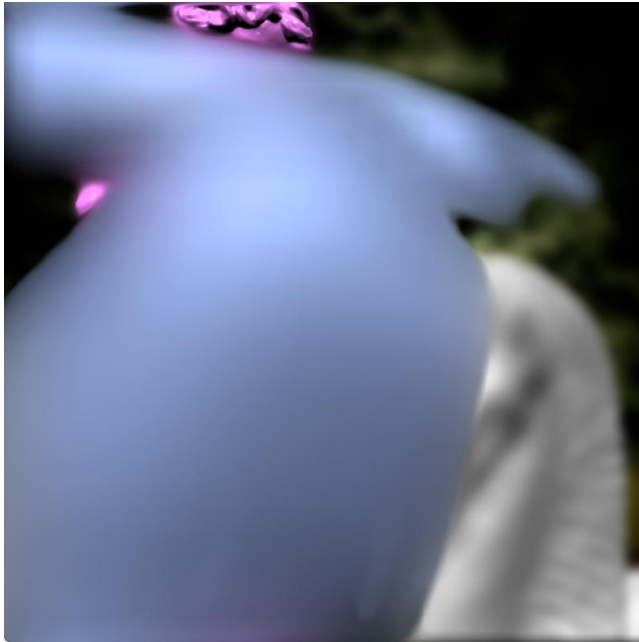
Starcraft 2 [Filion 2008]



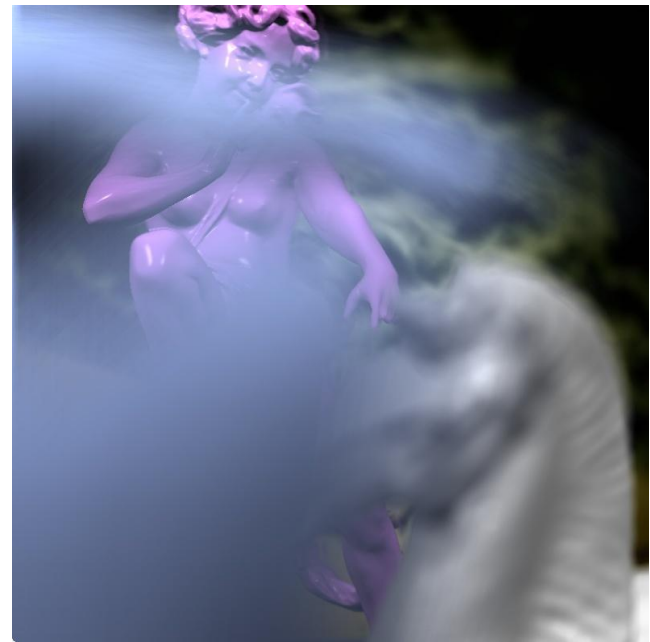
Limitations

- **Wrong visibilities**

- Single-view approximations using depth map cannot deliver correct visibility.



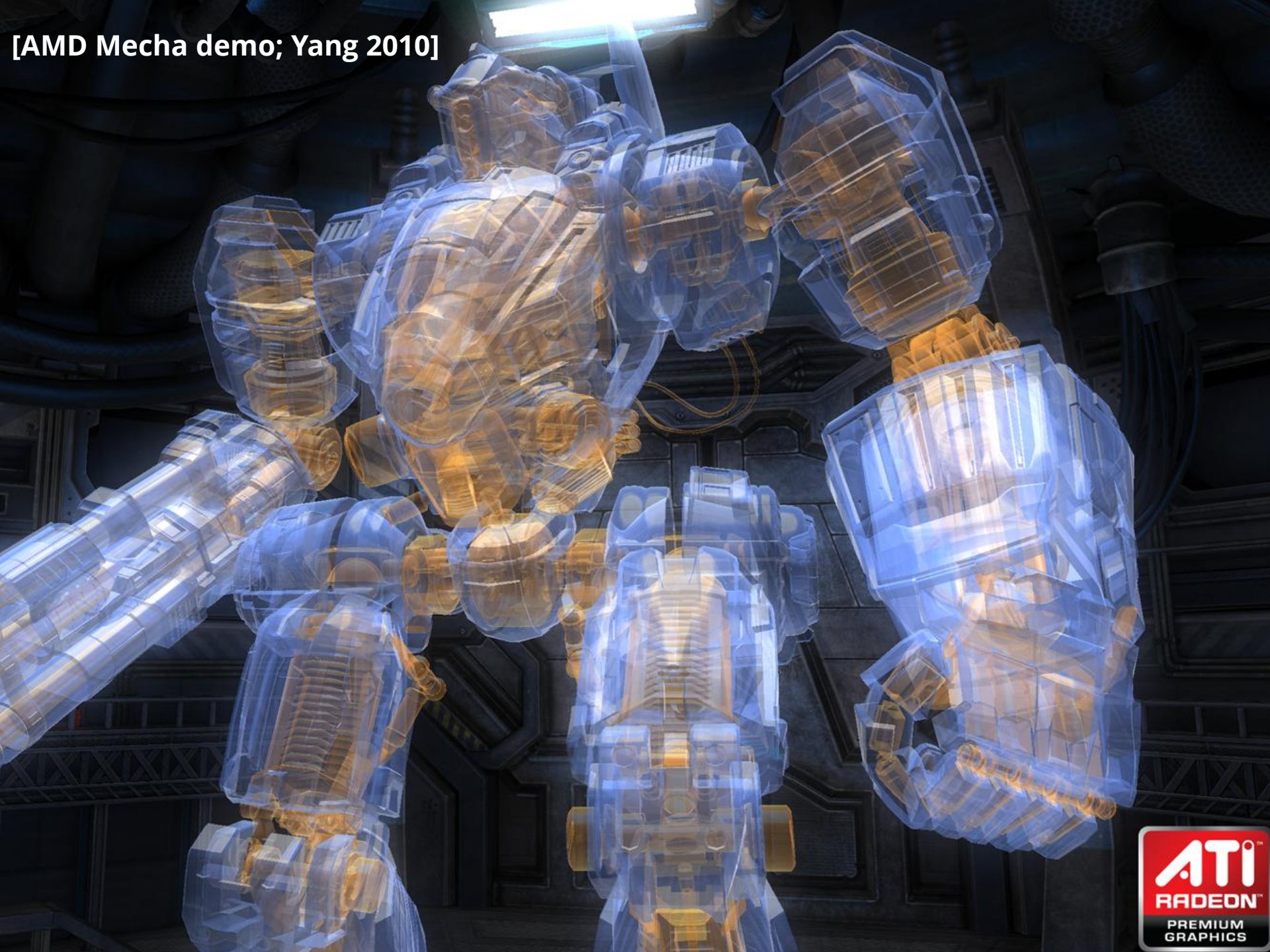
Typical foreground problem
(wrong partial occlusion)



Reference

Transparency

[AMD Mecha demo; Yang 2010]



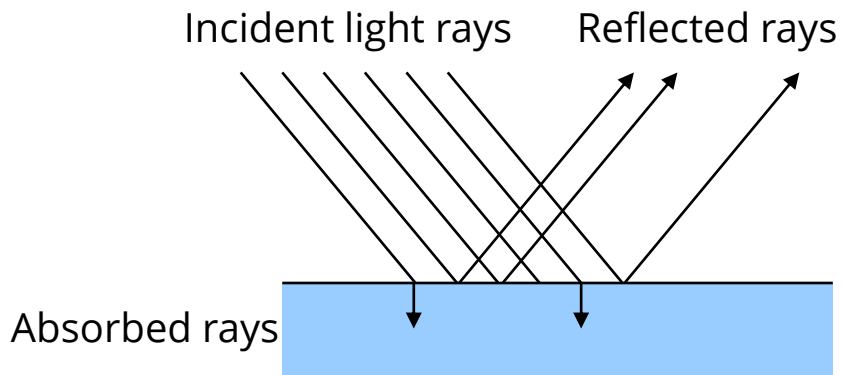
[McGuire 2013]



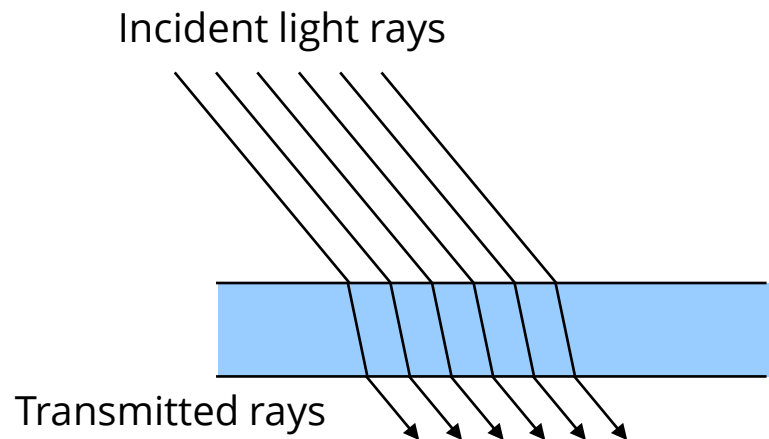
Transparency

- **Informal definition**

- The propagation of electromagnetic waves that pass through a transparent dielectric media.



Opaque object



Transparent object

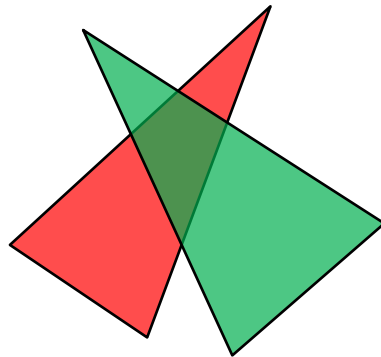
Transparency

- **Transparency uses alpha blending**

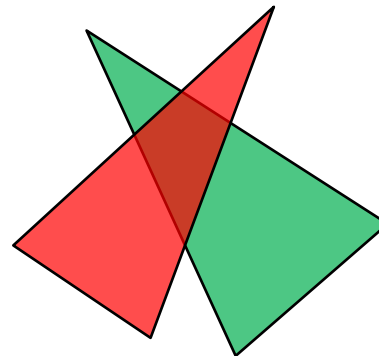
- Alpha channel encodes opacity.

- **Problems of alpha blending**

- The results of alpha blending rely on the order of fragments.
- We might end up with wrong visibility without proper order.
- Here, "*order*" means the rendering (or *blending*) order of geometries.



Front to back blending



Back to front blending

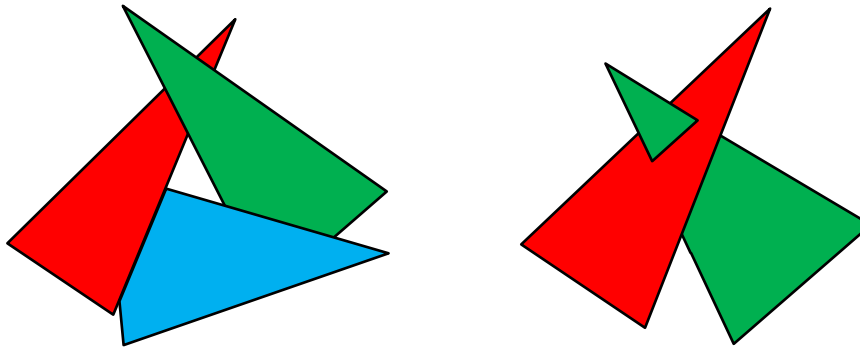
Order-Dependent Transparency (ODT)

- **Rendering**

- Render opaque objects first.
- Sort all transparent triangles before rendering (primitive sorting).
- Render transparent triangles with alpha blending

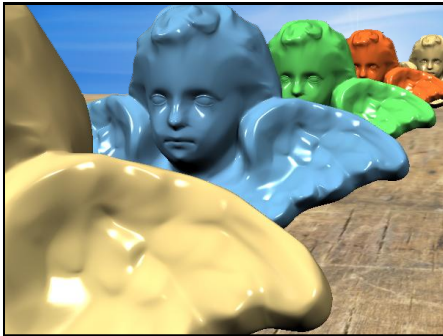
- **Limitations of order-dependent transparency**

- Depth sorting of triangles are complex or often impossible.



Order-Independent Transparency (OIT)

- **Interactive Order-Independent Transparency [Everitt 2001]**
 - General idea:
 - Triangles are rendered without depth sorting.
 - Instead, we apply depth sorting to visible pixels after rendering
 - Use **depth peeling** (depth sorting via layered rendering)
 - N geometry passes are need when the depth has N complexity.

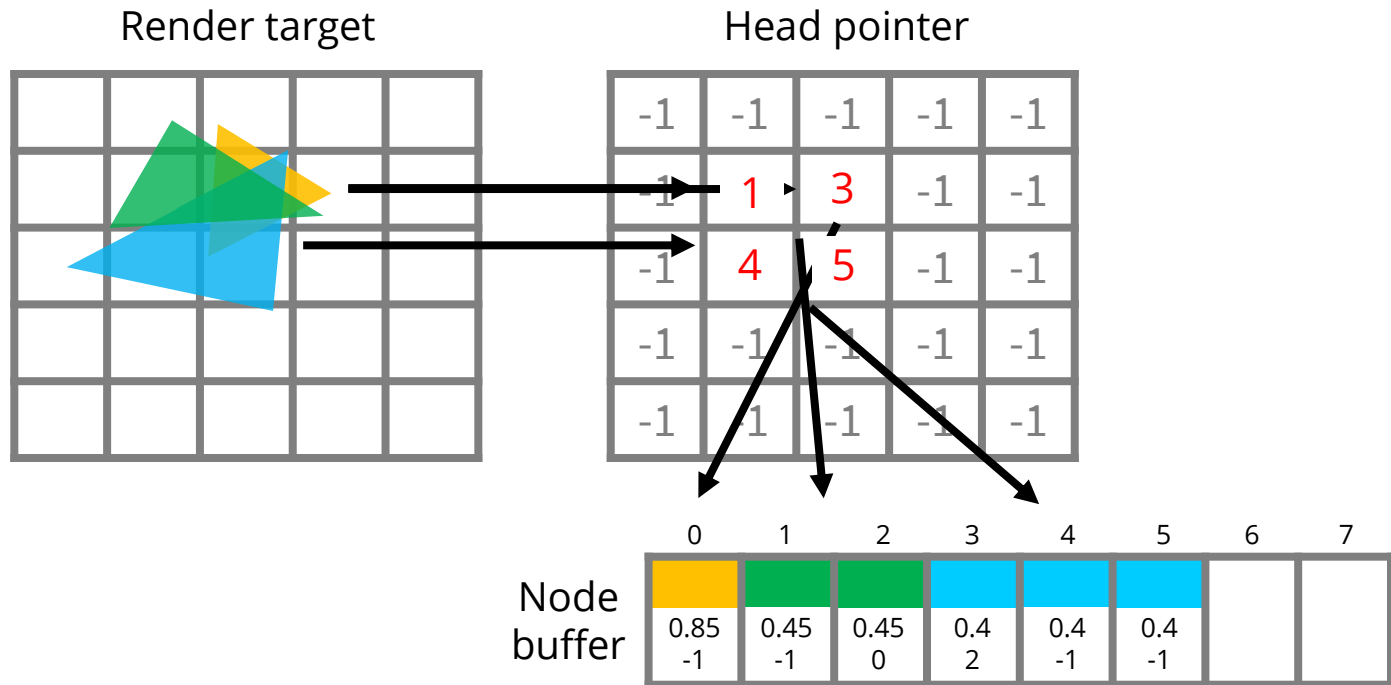


Order-Independent Transparency

- **Online fragment sorting with GPU linked lists [Yang 2011]**
 - All transparent fragments are stored using per-pixel linked lists.
 - Store fragments: color, alpha, and depth
 - A single geometry pass yet with an additional sorting pass

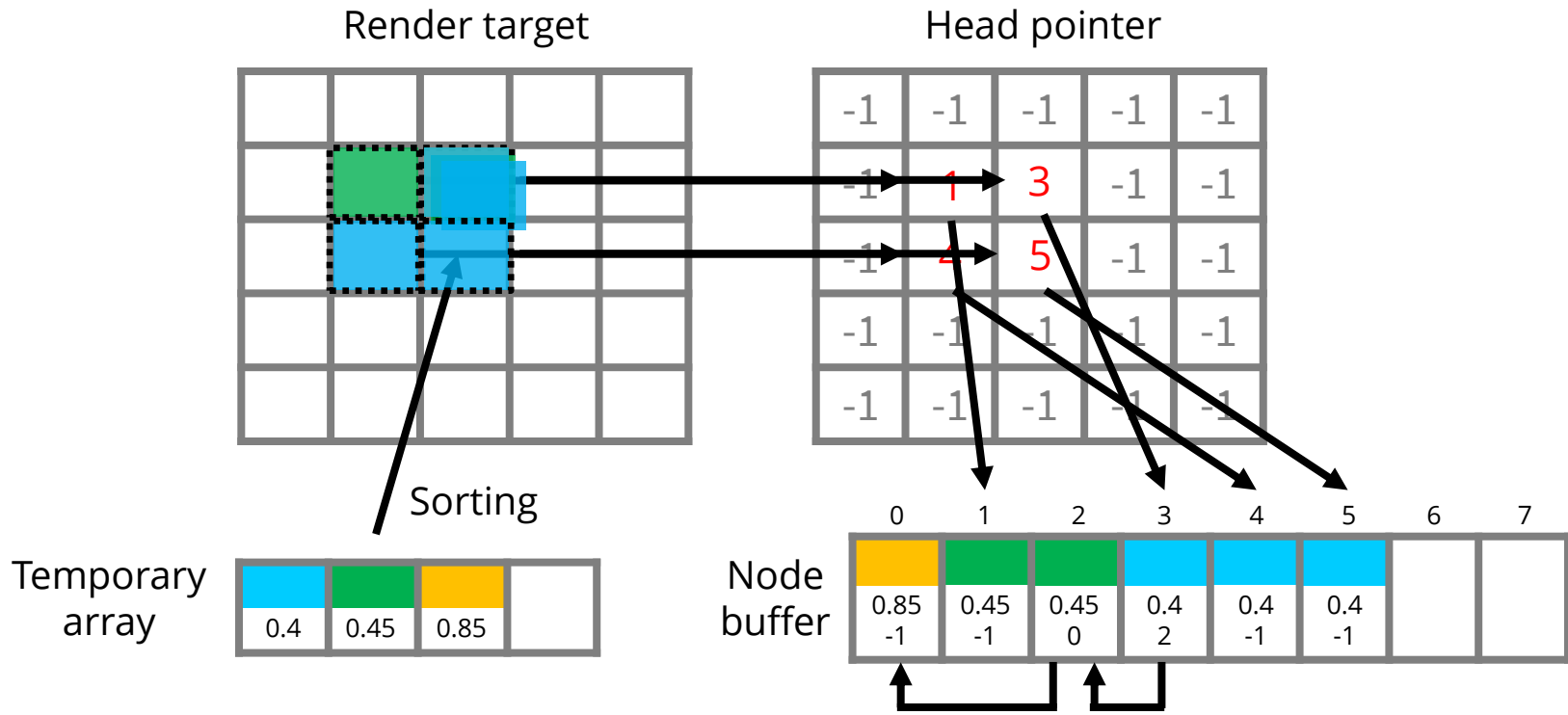
Order-Independent Transparency

- **Online fragment sorting with GPU linked lists [Yang 2011]**
 - First pass: per-pixel linked list construction in geometry rendering



Order-Independent Transparency

- **Online fragment sorting with GPU linked lists [Yang 2011]**
 - Second pass: fragment sorting



Bloom

Examples: High-Quality Bloom

- **Quad-Based Fourier Transform for Efficient Diffraction Synthesis (Scandolo et al., EGSR 2018)**

Realistic bloom rendering



Our approach can be used to compute a diffraction pattern to be used for realistic bloom rendering.

Bloom

- **Informal definition**

- Glow or halo by illumination when rays encounter obstacles.
- Fringes of light extending from the borders of bright areas in an image.
- Reproducing an imaging artifact of real-world cameras.

- **Benefits**

- Affect brightness of illumination or atmosphere.
- people accept the light with bloom effects is more brighter and stronger.

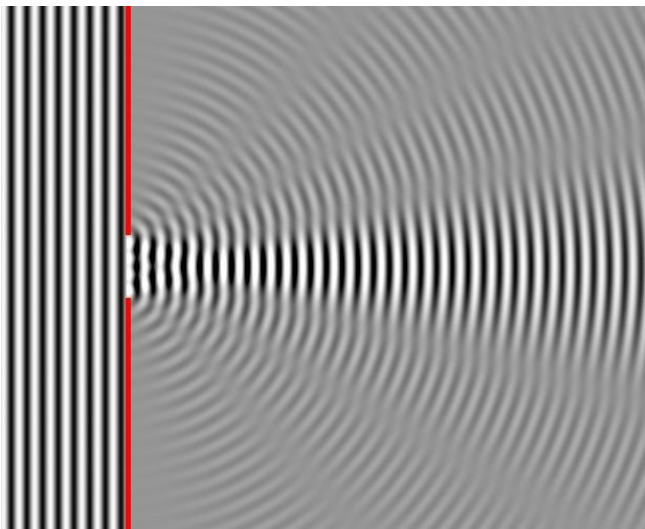


<http://support.nextlimit.com/display/maxwelldocs/Lens+Scattering>

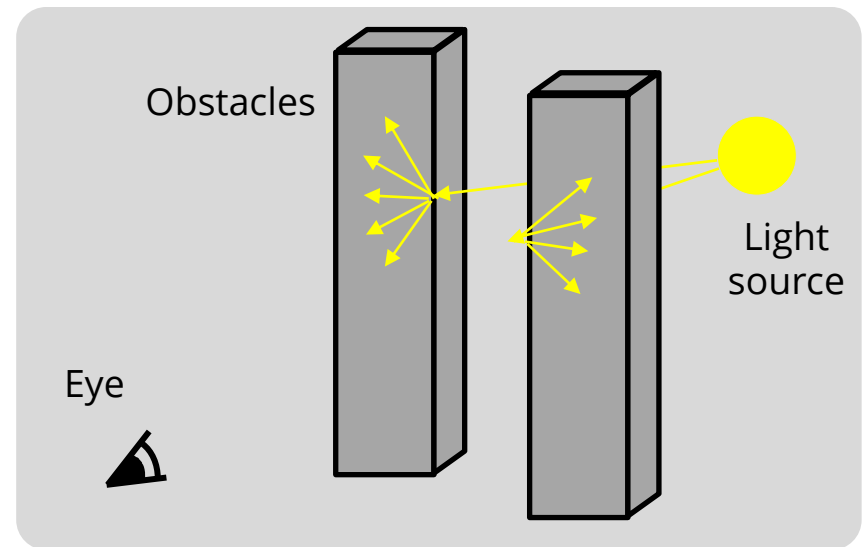
Physics

- **Underlying physics:**

- When light encounters an obstacle or a slit, diffraction effects occur.
- Intense light source makes the diffraction effects visible.
- The image of the bright lights appears to spread out near the borders of obstacles.



Diffraction pattern, Wikipedia

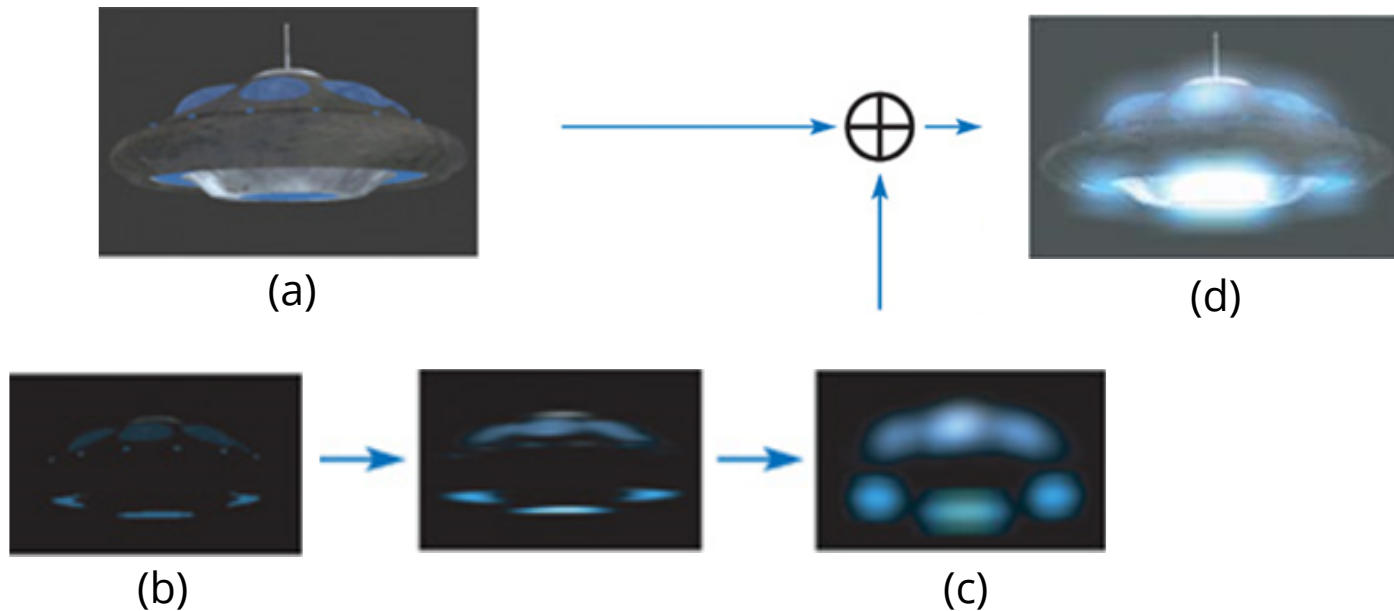


Spreading lights

Algorithm

- **GPU Gems, NVIDIA**

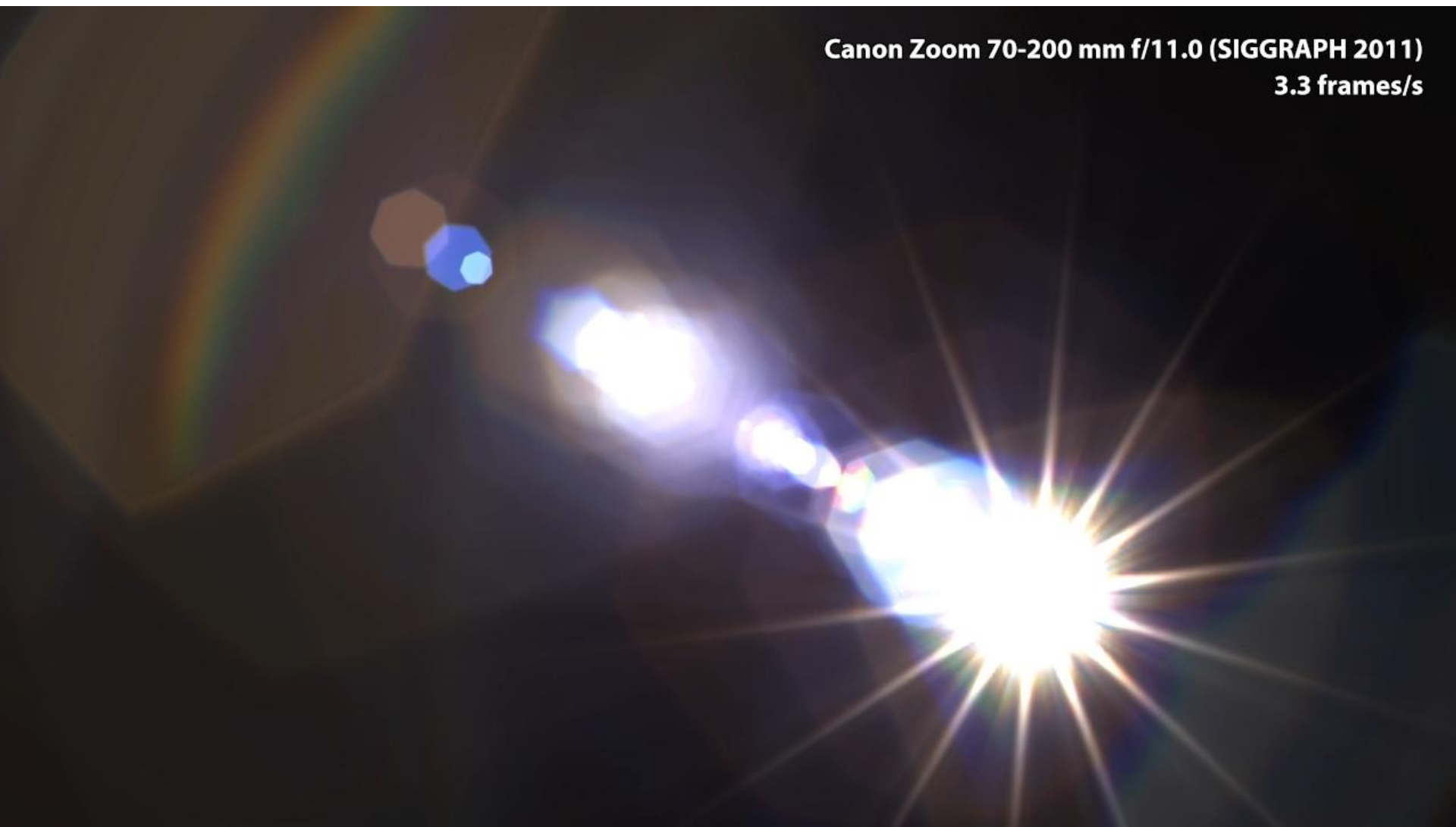
- Isolate non glowing parts and glowing parts (b).
- Blur the glowing objects (c).
- Compose the normal scene and the blurred scene (d).



Lens Flares

Examples

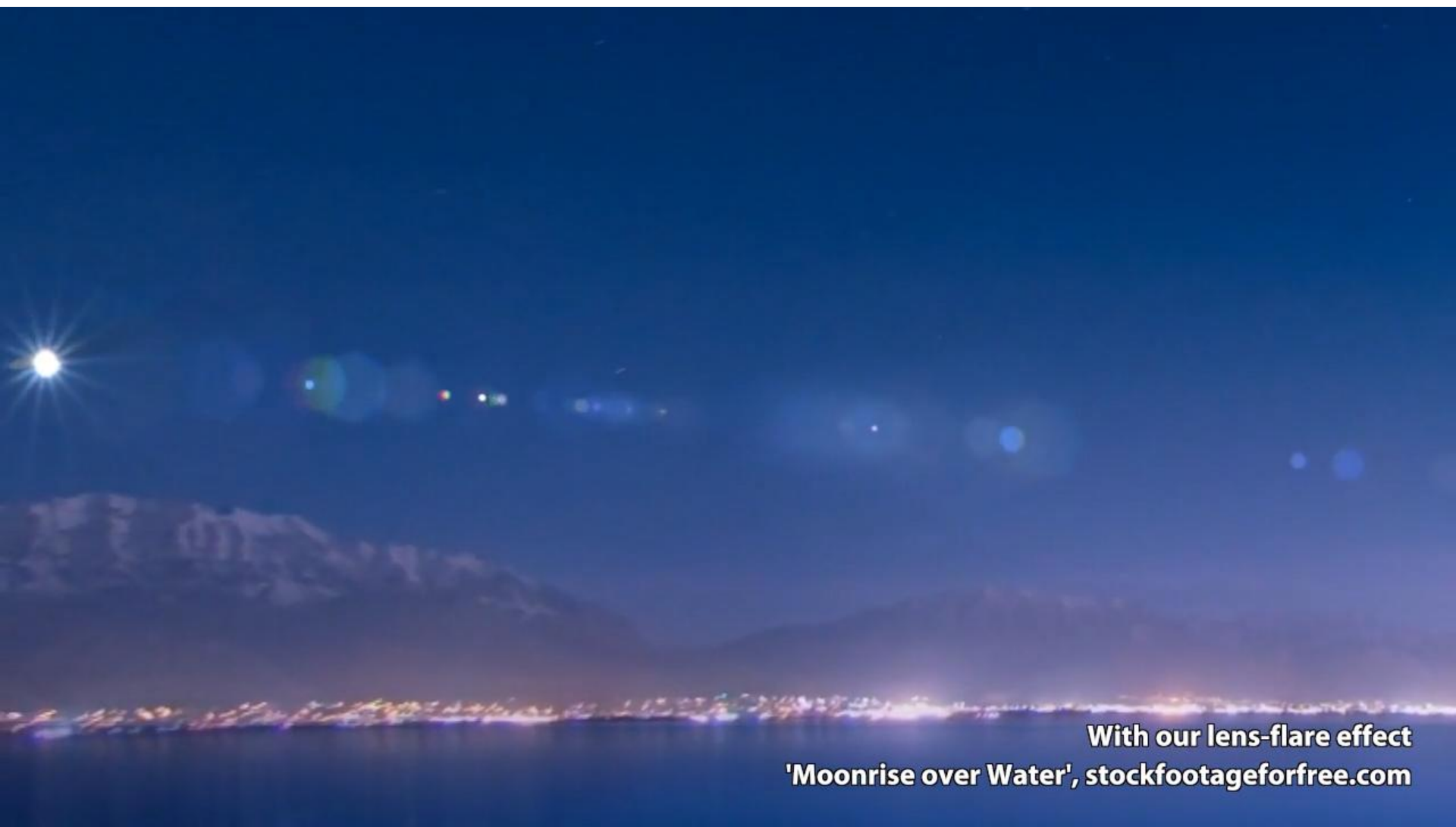
- **Physically-Based Real-Time Lens Flare Rendering (Hullin et al., SIGGRAPH 2011)**



Canon Zoom 70-200 mm f/11.0 (SIGGRAPH 2011)
3.3 frames/s

Examples

- **Practical Lens Flare Rendering (Lee and Eisemann, EGSR 2013)**



With our lens-flare effect
'Moonrise over Water', stockfootageforfree.com

Lens Flare

- **Informal definition**

- Unintended excessive light caused by refraction, reflection and scattering when light pass through a photographic lens system.

- **Benefits**

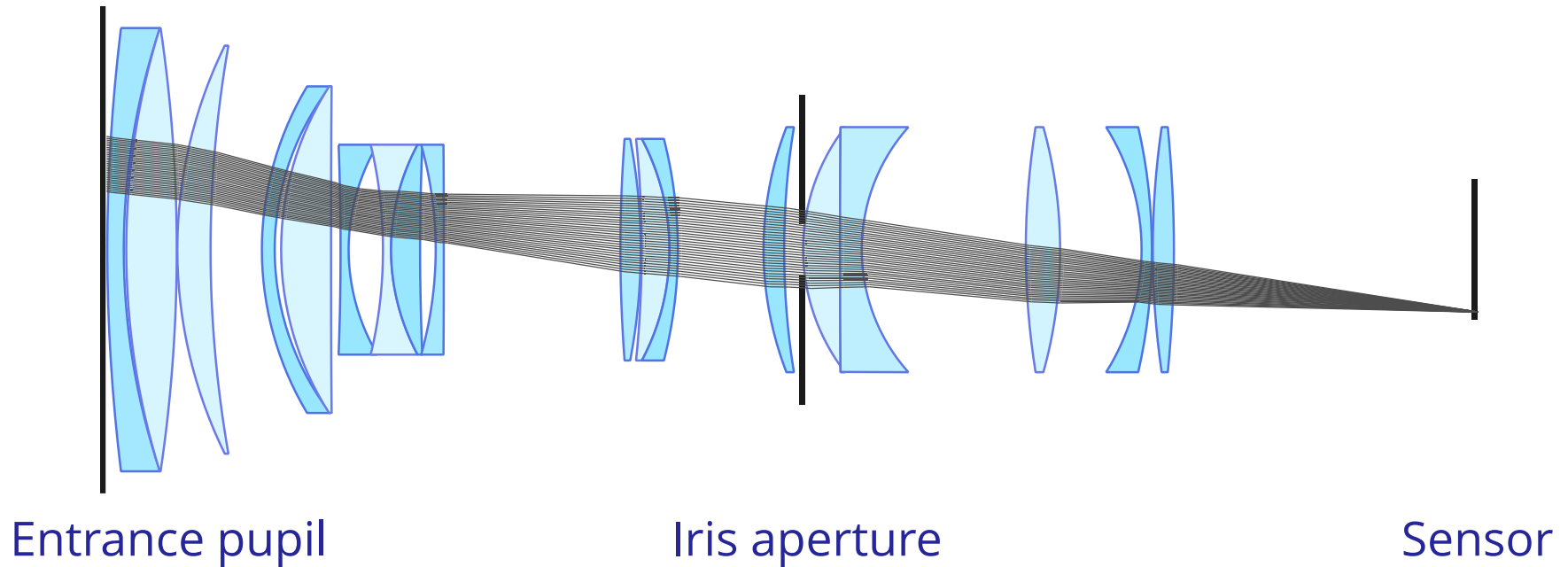
- Suggest the light sources are very bright.
- Increase the perceived realism.



Physics

- **Underlying physics:**

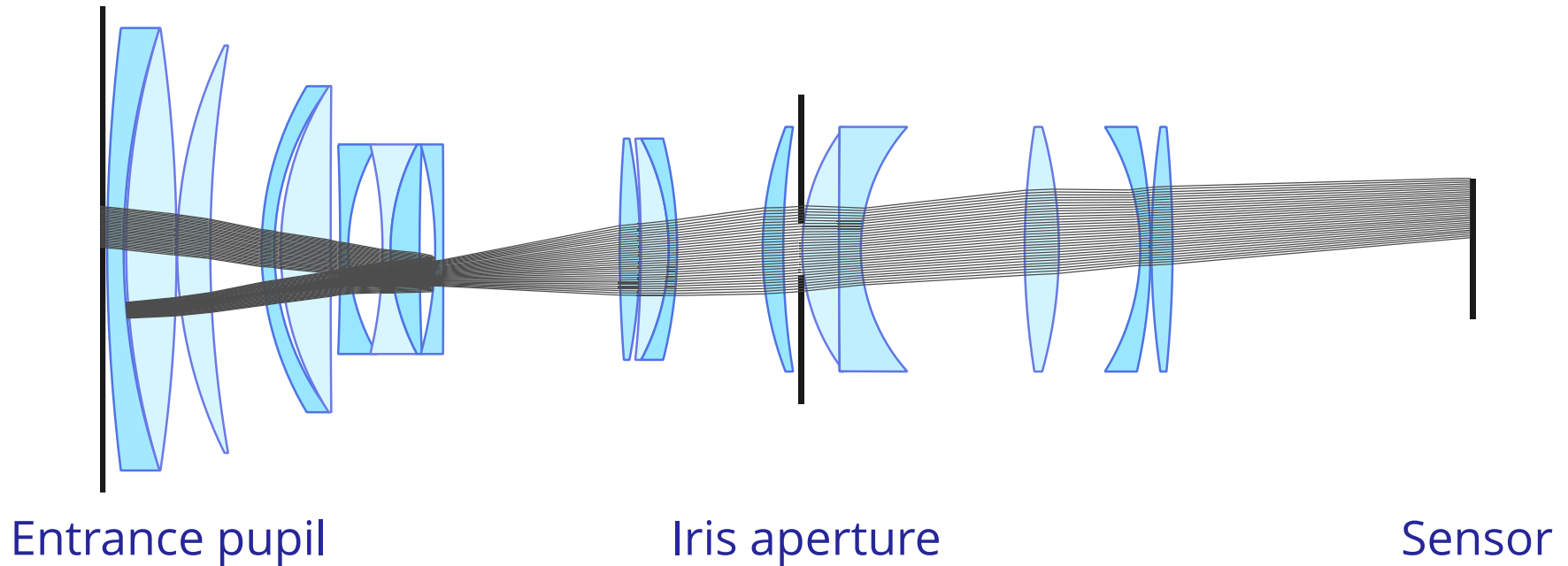
- In optical systems, a set of lenses are supposed to **refract** an incoming light ray so that it reaches the sensor.



Physics

- **Underlying physics:**

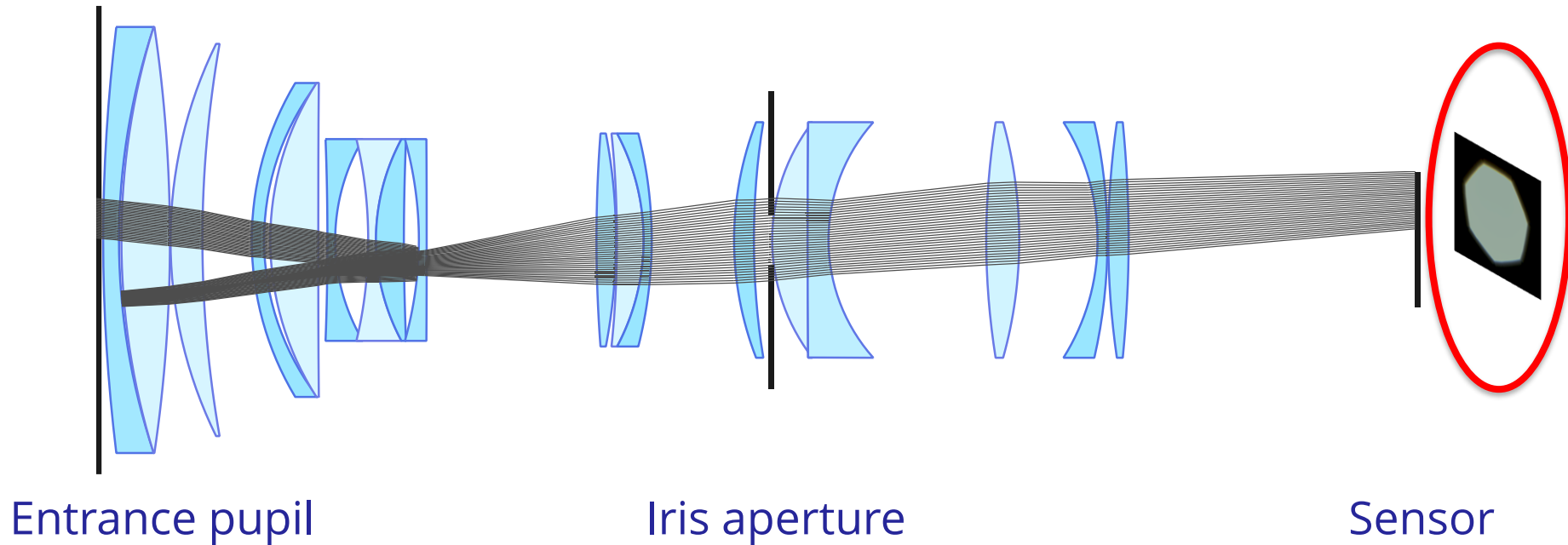
- However, though manufacturers unwanted, **lenses always reflect** small fraction of light, resulting in deviations from the intended trajectories.



Physics

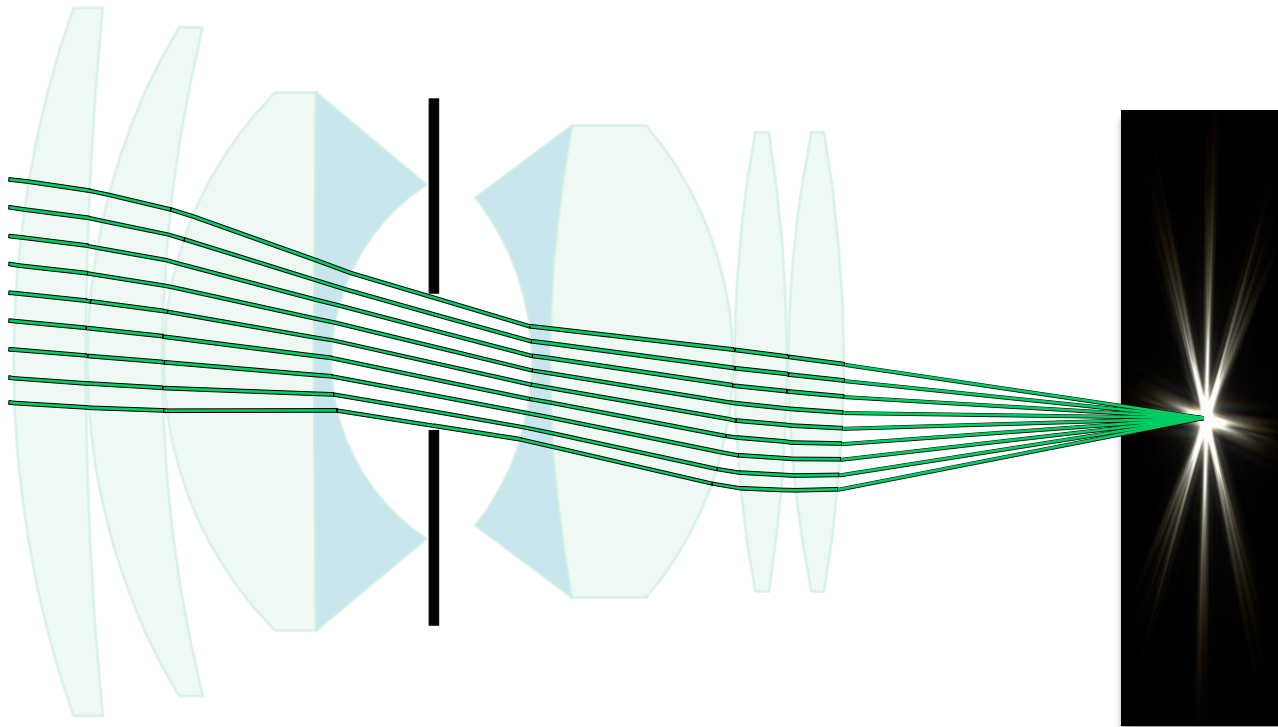
- **Underlying physics:**

- We can observe the reflection patterns, so called the "**ghosts**"
- As the rays still cross the aperture, the ghosts share the aperture shape.



References

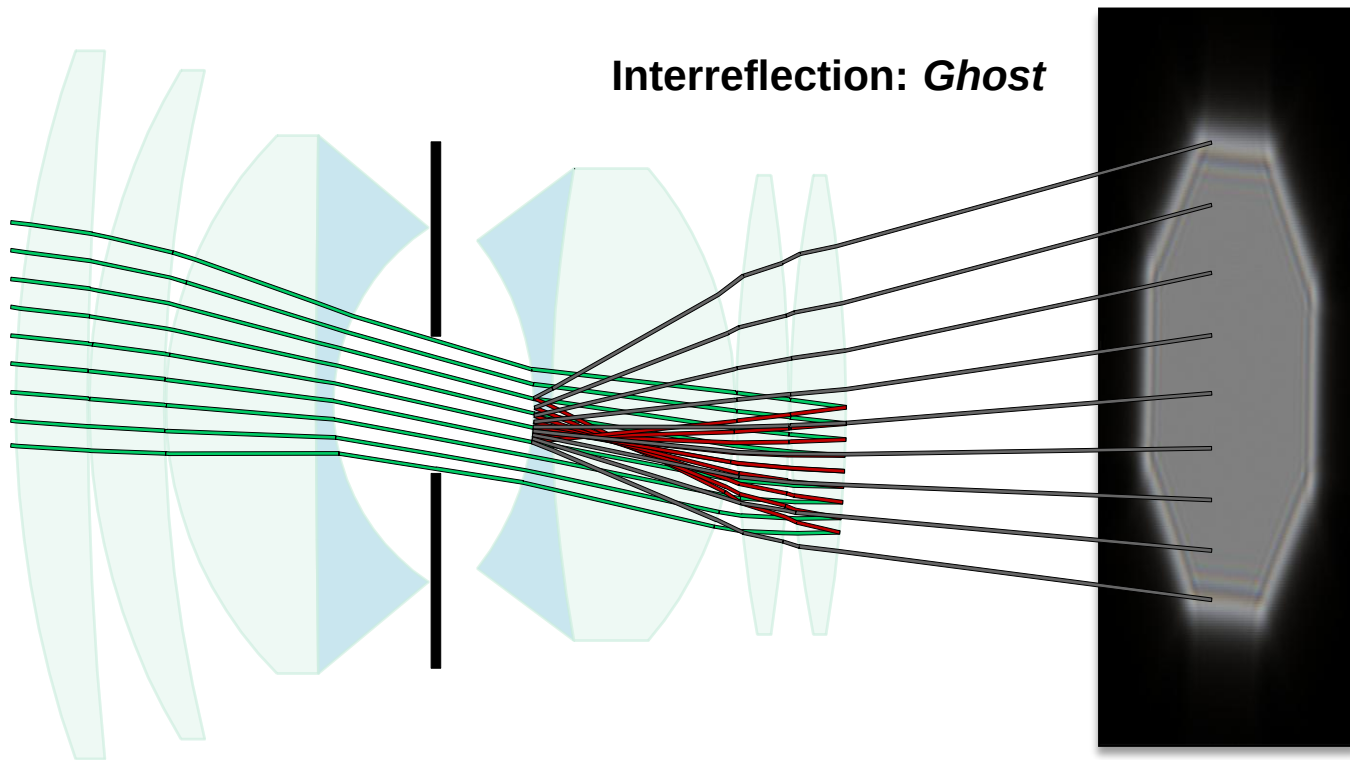
- **Physically-Based Real-Time Lens Flare Rendering [Lee 2011]**
 - Make a model of the optical system.
 - Simulate ghost effects and a starburst effect.



Direct light: *Starburst*

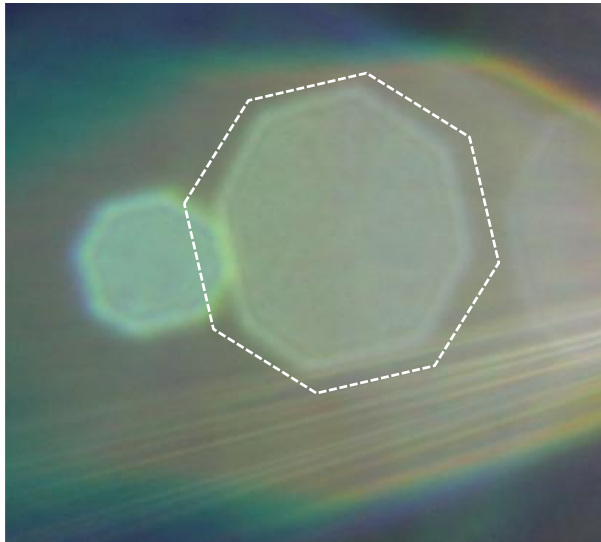
References

- **Physically-Based Real-Time Lens Flare Rendering [Lee 2011]**
 - Make a model of the optical system.
 - Simulate ghost effects and a starburst effect through ray tracing.

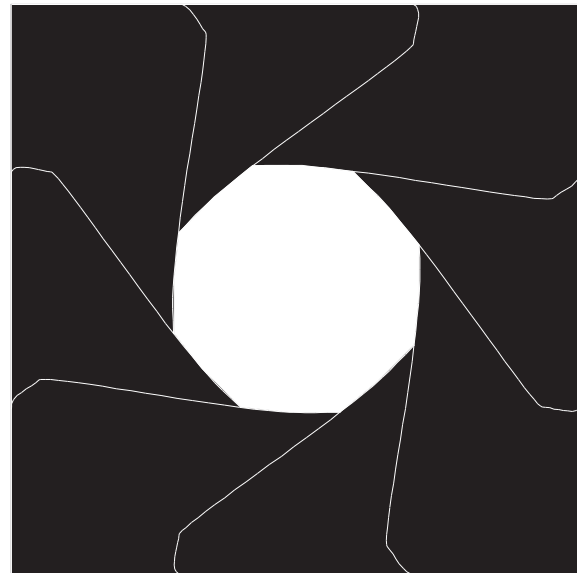


References

- **Physically-Based Real-Time Lens Flare Rendering [Lee 2011]**
 - Ghost effects
 - Decide the shape of ghosts through the shape of an aperture.



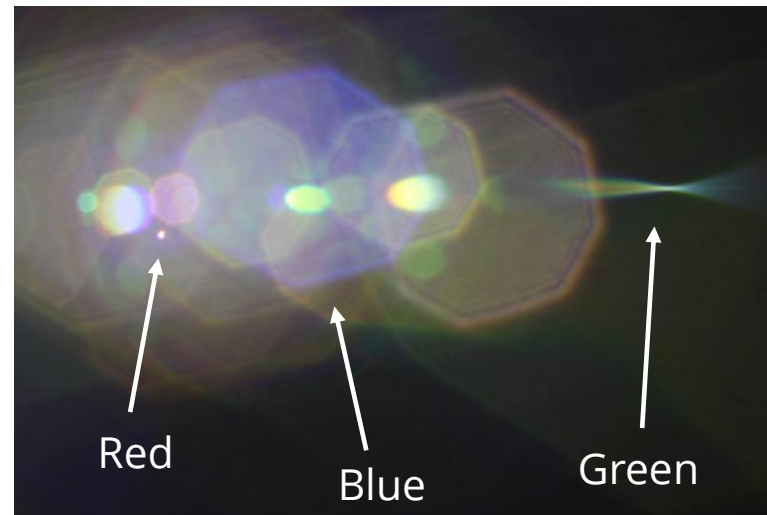
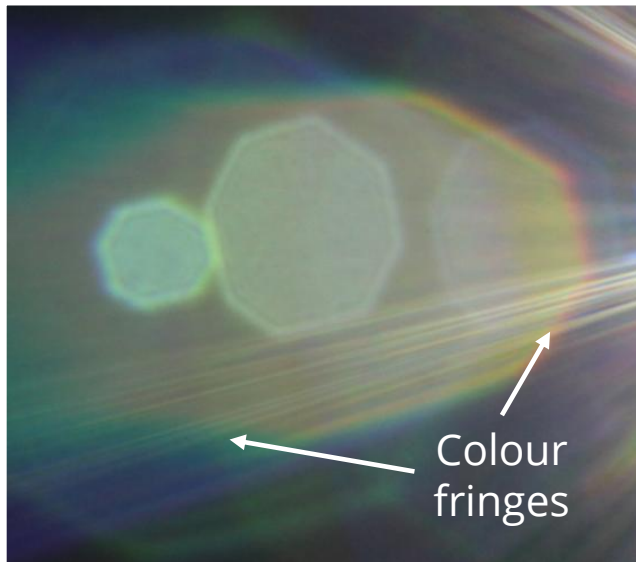
Shape of ghosts



Shape of an aperture

References

- **Physically-Based Real-Time Lens Flare Rendering [Lee 2011]**
 - Ghost effects
 - Decide the shape of ghosts through the shape of an aperture.
 - Simulate dispersion or reflectivity to present features of the ghost such as color fringes and diverse colors .



References

- **Physically-Based Real-Time Lens Flare Rendering [Lee 2011]**
 - Starburst effects
 - Get a diffraction pattern of the aperture using Fraunhofer approximation.

